

Lab # 06

Working with Subqueries in Oracle Database



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Database Systems Lab (CL-2005)

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Working with Subqueries in Oracle Database

Objectives:

- To understand the concept of subqueries in Oracle Database.
- To learn different types of subqueries.
- To implement single-row, multiple-row, correlated, and DML subqueries using the HR schema.
- To understand how subqueries are executed in Oracle SQL.

Introduction to Subqueries

A **subquery** is a SQL query written inside another SQL statement. It is also referred to as:

- **Inner Query** – the nested query
- **Outer Query** – the main query

In Oracle Database, subqueries can be used within:

- SELECT statements
- INSERT statements
- UPDATE statements
- DELETE statements
- WHERE clause
- HAVING clause
- FROM clause (Inline View)

The inner query is executed first. Its result is then passed to the outer query for further processing. Subqueries are particularly useful when a condition depends on a value that is not directly known beforehand.

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General Syntax of Subquery in Oracle:

```
SELECT column_name
FROM table_name
WHERE column_name operator
      (SELECT column_name
       FROM table_name
       WHERE condition);
```

Execution Process:

- The inner query executes first.
- It returns a result (single value or multiple values).
- The outer query uses that result to filter or compare records.

Rules and Guidelines for Subqueries

When working with subqueries in Oracle Database, the following rules must be observed:

- A subquery must always be enclosed in parentheses ().
- The subquery appears on the right-hand side of a comparison operator.
- Single-row operators must be used with single-row subqueries.
- Multiple-row operators must be used with multiple-row subqueries.
- The ORDER BY clause is not allowed in simple subqueries.
- If a single-row subquery returns more than one row, Oracle generates the error:
ORA-01427: single-row subquery returns more than one row
- A subquery may return:
 - A single value (scalar)
 - A single row
 - Multiple rows
 - Multiple columns

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Types of Subqueries in Oracle

The main types of subqueries covered in this lab are:

- Single Row Subquery
- Multiple Row Subquery
- Correlated Subquery
- Subqueries using IN, ANY, and ALL
- Row Subqueries
- EXISTS and NOT EXISTS
- Subqueries in the FROM clause (Inline Views)
- Subqueries with DML statements

Single Row Subquery

- A single-row subquery returns exactly one row and one column.
- It is used with single-row comparison operators such as: =, >, >=, <, <=, <>
- If the subquery returns more than one row, Oracle generates an error.

Example: Employee with Highest Salary

```
SELECT employee_id, first_name, last_name, salary
FROM employees
WHERE salary =
  (SELECT MAX(salary)
   FROM employees);
```

Explanation:

- The inner query returns the maximum salary from the employees table.
- The outer query retrieves the employee(s) whose salary matches that value.

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Example: Employees Earning More Than Average Salary

```
SELECT employee_id, first_name, last_name, salary
FROM employees
WHERE salary >
  (SELECT AVG(salary)
   FROM employees);
```

The subquery calculates the average salary, and the outer query compares each employee's salary with that value.

Multiple Row Subquery

- A multiple-row subquery returns more than one row.
- Single-row operators ($=$, $>$, $<$) cannot be used directly with such subqueries. Instead, Oracle provides: IN, ANY, ALL

IN Operator

The IN operator returns TRUE if the value matches any value returned by the subquery.

```
SELECT first_name, department_id
FROM employees
WHERE department_id IN
  (SELECT department_id
   FROM departments
   WHERE location_id = 1700);
```

The subquery returns multiple department IDs, and the outer query retrieves employees working in any of those departments.

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ANY Operator

The ANY operator returns TRUE if the comparison condition is satisfied for at least one value returned by the subquery.

```
SELECT employee_id, first_name, salary
FROM employees
WHERE salary < ANY
  (SELECT salary
   FROM employees
   WHERE job_id = 'SH_CLERK');
```

This query retrieves employees whose salary is less than at least one salary of a Shipping Clerk.

ALL Operator

The ALL operator returns TRUE only if the condition is satisfied for all values returned by the subquery.

```
SELECT employee_id, first_name, salary
FROM employees
WHERE salary > ALL
  (SELECT salary
   FROM employees
   WHERE job_id = 'SH_CLERK');
```

This query retrieves employees earning more than the highest Shipping Clerk salary.

Correlated Subquery

- A correlated subquery refers to a column from the outer query.
- Unlike simple subqueries, a correlated subquery executes once for each row processed by the outer query.

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Example: Employees Earning More Than Department Average

```
SELECT employee_id, first_name, salary, department_id
FROM employees e
WHERE salary >
    (SELECT AVG(salary)
     FROM employees
     WHERE department_id = e.department_id);
```

For each employee, the subquery calculates the average salary of that employee's department. The comparison is then performed individually for each row.

Row Subquery

- A row subquery returns one row containing multiple columns.
- The number of columns in the outer query condition must match the number of columns returned by the subquery.

Example:

```
SELECT first_name
FROM employees
WHERE (department_id, manager_id) =
    (SELECT department_id, manager_id
     FROM departments
     WHERE location_id = 1800);
```

Both sides of the comparison contain two columns.

Subqueries in the FROM Clause (Inline Views)

- A subquery placed inside the FROM clause is known as an **Inline View**.
- It acts like a temporary table.
- An alias is mandatory.

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Example:

```
SELECT department_id, avg_salary
FROM (
    SELECT department_id, AVG(salary) AS avg_salary
    FROM employees
    GROUP BY department_id
)
WHERE avg_salary > 8000;
```

The inner query creates a temporary result set which is then filtered by the outer query.

Subqueries with DML Statements

Subqueries can also be used with INSERT, UPDATE, and DELETE statements.

Subquery with INSERT

```
INSERT INTO employee_bkp
SELECT *
FROM employees
WHERE job_id IN
    (SELECT job_id
    FROM jobs
    WHERE job_title = 'Accountant');
```

Subquery with UPDATE

```
UPDATE employees
SET salary = salary + (salary * 0.10)
WHERE job_id IN
    (SELECT job_id
    FROM jobs
    WHERE min_salary = 3000);
```

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Subquery with DELETE

```
DELETE FROM employee_bkp
WHERE job_id IN
  (SELECT job_id
   FROM job_history
   WHERE end_date = TO_DATE('31-DEC-2006','DD-MON-YYYY'));
```

Exists and Non Exists Operators in Subqueries

Subquery using EXISTS

- The **EXISTS** operator checks whether the subquery returns at least one row.
- It is commonly used with correlated subqueries.

Example: Employees Who Are Managers

```
SELECT employee_id, first_name, last_name
FROM employees e
WHERE EXISTS
  (SELECT 1
   FROM employees
   WHERE manager_id = e.employee_id);
```

If at least one employee reports to them, the condition becomes TRUE.

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Subquery using NON EXISTS

The NOT EXISTS operator returns TRUE if the subquery returns no rows.

Example:

```
SELECT department_id, department_name
FROM departments d
WHERE NOT EXISTS
  (SELECT 1
   FROM employees
   WHERE department_id = d.department_id);
```

This query retrieves departments that have no employees.

THE END!